MILITARY SPECIFICATION

METERS, MILLIAMMETERS, DIRECT CURRENT PANEL

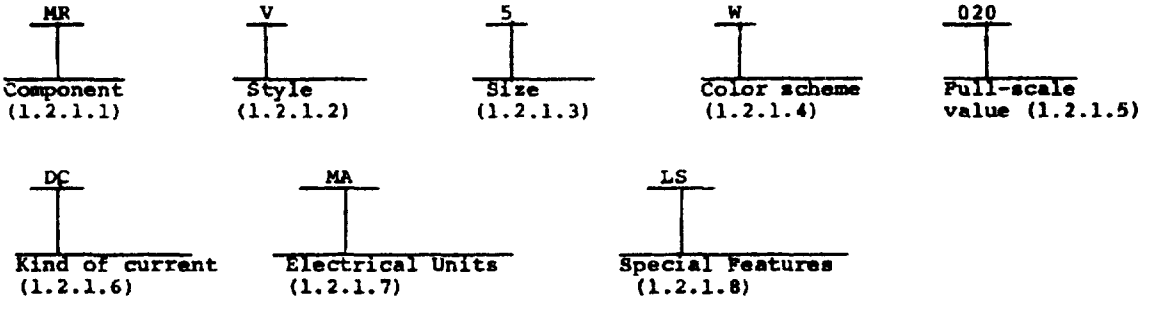
MOUNTING (EDGEWISE TYPES)

1. SCOPE

1.1 Scope. This specification covers panel mounted, electrical-indicating, milliampere, direct-current ammeters of the edgewise type.

1.2 Classification.

1.2.1 Type designation. The type designation of ammeters shall be in the following form, and as specified (see 6.1):



1.2.1.1 Component. Ammeters covered by this specification are designated by the two-letter symbol "MR".

1.2.1.2 Style. The style is designated by a one-letter symbol as follows:

V - vertical scale
H - horizontal scale

1.2.1.3 Size. The size is designated by a one-number symbol representing the conventional identification and approximately equal to the case size in inches as follows.

5 - 5-inch case length
8 - 8-inch case length

1.2.1.4 Color scheme. The scale color scheme is designated by a one-letter symbol as follows:

W - white scale background, black markings
B - black scale background, white translucent markings

1.2.1.5 Full-scale value.

1.2.1.5.1 Single-range meters. The full-scale value of single-range meters with zero at left for horizontal scale meters or at the bottom for vertical scale meters is represented by three figures designating the maximum number of units indicated; when the full-scale value is less than three figures, zeros are inserted at the left to fill out to three figures. Where the letter "R" is used in this group, between two figures, it represents a decimal point. Example: 1R5 represents 1.5.

1.2.1.5.2 Offset zero. The full-scale value of zero-center or offset zero meters is indicated by two numbers designating the numerical value of the end-scale points, with a letter between them designating the decimal value of the numbers, as follows:

D - tenths Example: 5D5 represents 0.5-0-0.5.
U - units Example: 1U9 represents 1-0-9.
T - tens Example: 2T2 represents 20-0-20.
H - hundreds Example 5H5 represents 500-0-500.

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

1.2.1.6 Kind of current. The kind of current for which the ammeters are designed is designated by two letters as follows:

DC - Direct current

1.2.1.6.1 Standard ranges. Ammeters shall be furnished in the following full-scale ranges

Direct current ammeters
Milliamperes

1 10 100

2 20 200

5 50 500

1.2.1.7 Electrical units. Two letters that designate the electrical units indicated by the ammeters are MA-Milliamperes.

1.2.1.8 Special features. A two letter symbol designates special features as follows. An ammeter may have no special feature or one or two special features.

1st letter
O - no illumination
L - Internal illumination
2nd letter
O - no special marking
S - special scale marking, for example a non-electrical quantity such as gallons per minute or pounds per square inch.

1.2.1.9 Examples of type designations. Type MRV5W050DCMA00 signifies·

(a) MR - Ammeter
(b) V - vertical
(c) 5 - 5-inch case size
(d) W - White scale
(e) 050 - Zero bottom, full-scale value of 50.
(f) DC - direct current
(g) MA - Milliampere
(h) 0 - no illumination
(i) 0 - no special marking

Type MRH8W005DCMALS signifies

(a) MR - Ammeter
(b) H - horizontal
(c) 8 - 8-inch case size
(d) W - White scale
(e) 005 - Zero left, full-scale value of 5
(f) DC - Direct current
(g) MA - Milliampere
(h) L - internal illumination
(i) S - special scale, as specified (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery Equipment and Systems
MIL-E-917 - Electric Power Equipment Basic Requirements for (Naval Shipboard).

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MS 25237 - Lamp, Incandescent, Single Contact Midget Flanged Base (T-1-3/4 Bulb).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

UNITED STATES OF AMERICAN STANDARDS INSTITUTE
USAS C39.1-1964 - Electrical Indicating Instruments.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Sample for First Article inspection. Prior to beginning production, samples shall be tested as specified in 4.3 (see 6.1).

3.2 Definitions. Definitions of terms used in this specification shall be those definitions used in United States of America Standards Institute (USAS) C39.1-1964 and herein.

3.2.1 Friction effect. The friction effect shall be determined as follows: Untapped ascending and descending readings shall be taken at five points distributed over the scale, using the same scale points for ascending and descending readings. The difference between corresponding ascending and descending readings as observed on the standard meter shall be determined.

3.3 Material. Material requirements of MIL-E-917 shall apply. Thermoplastic materials may be used for minor parts associated with internal illumination features where used. Total temperature of thermoplastic materials so used shall not exceed 85°C at 65°C ambient with meter and internal illumination circuit energized.

3.4 Finish. The portion of the ammeter case exposed to view from the front of the panel shall have a light gray finish having a Munsell rating of 2.5GY6.0/0.5. No nickel or other bright trimmings shall be used. Metal cases shall be rendered resistant to corrosion prior to the application of the final gray finish. Pretreatment and painting shall conform to MIL-E-917.

3.5 Design and construction.

3.5.1 Balancing of moving element. The moving element of the ammeters shall be balanced by suitable means such as threaded locknuts or splitnuts, or a wire helix moving along a crossarm and counterweight arm. The use of solder, or an adhesive such as shellac, to hold the balance weights or to function as a part of the balance weight, shall not be used except where the ammeters has a separate non-soldered adjustable balance weight.

Source: https://assist.dla.mil -- Downloaded: 202640-19T21:31Z
Check the source to verify that this is the current version before use.

3.5.2 Magnets. If laminated or multiple-piece magnets are used, the laminations or the parts shall be securely fastened together so that they will not separate or loosen as a result of tests specified herein.

3.5.3 Cases. The ammeter cases shall be made of molded thermosetting plastic material, metal, or other material as specified herein. The cases shall be designed and constructed with close-fitting joints to minimize the entrance of dust and moisture.

3.5.4 Window. Ammeters shall be provided with a window of a methy methacrylate. The windows shall be free from detrimental defects that would prevent the ammeter from being easily read. Such defects include electrostatic effects, scratches, chips, cracks, striae, or craze. This requirement applies both before and after the tests specified herein.

3.5.5 Sealing. The ammeter cases shall be sealed by means of gaskets, by fusing or soldering metal-to-metal, or metal-to-glass, or by any other means which will enable the ammeter to withstand the tests specified herein.

3.5.6 Scale and pointer.

3.5.6.1 Scale construction. The scale shall be made of stiff material firmly supported and secured to its mounting. Paper scales will be satisfactory only when adequately bonded to and backed by a metal support. The scale shall be of such material and so secured as to insure freedom from warping, fading, or other deterioration during the tests specified herein. The scale shall be bilevel (formed so that the scale divisions lie in the same plane as the pointer). The quantity being measured shall be printed on the scale level which is recessed so that the pointer can swing over it. The divisions and numbers (for example, 0, 1, 2, and so forth) shall be printed on the other scale level. The color scheme for the dial shall be either white scale background with black markings, or black scale background with white translucent markings (see 1.2.1.4).

3.5.6.2 Scale length and marking.

3.5.6.2.1 Length. The minimum scale length shall be 4-inches for the 5-inch ammeter and 6.3 inches for 8-inch ammeter.

3.5.6.2.2 Divisions. The value of each scale divisions shall, whenever practicable, be either 1, 2, or 5 of the units measured or any decimal multiple or submultiple of these numbers. In the case of multiple scale ammeters or of ammeters for other specialized uses, exceptions to this requirement may be necessary, but shall be avoided when reasonably possible. The angle subtended by a scale division shall be not less than 1 degree.

3.5.6.2.3 Character size. Numbers and letters for scale marking shall be 3/16 inch high, minimum for 5-inch meters and 1/4 inch, minimum for 8-inch meters. Style shall be number 607 Twentieth Century Bold Gothic Type.

3.5.6.3 Pointer. The ammeters shall have a pointer so formed as to permit accurate readings at the usual distance from which the ammeter is viewed, and approximate readings up to the distance of legibility of the scale markings. The pointer shall be of rigid construction and suitable for the intended application and shall not become damaged or distorted when subjected to any of the tests specified herein. The pointer shall extend over the edge of the scale plate at least 3/8 inch (0.375 inch) for 5 inch-meters and 1/2-inch (0.5 inch) for 8-inch meters.

3.5.7 Means of adjustment.

3.5.7.1 Accuracy adjustment. Ammeters shall be provided with practical shop means of adjustment to accuracy. Charging of the magnet and magnetically treating it to obtain accuracy, or adjusting the internal resistor, shall be considered as meeting this requirement.

3.5.7.2 Zero adjuster. Each ammeter shall have a zero adjuster accessible from the front of the case. This zero adjuster shall be a screw type capable of 360 degrees rotation, or a sliding tab protruding from the ammeter front alongside the window. If the sliding tab method is used, the tab shall be recessed so that it does not extend beyond the ammeter front. The tab shall slide between two rubber or plastic lips which effectively exclude drops of water when the ammeter is tested as specified in 4.6.4.

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

3.5.8 Terminals and connections.

3.5.8.1 Terminals. Terminals shall be threaded stud 1/4-28 type in accordance with Handbook H28 (see 3.11). The terminals shall be anchored to the rear of the ammeter case in a manner which prevents turning. Two loose nuts and two washers shall be furnished on each terminal.

3.5.8.1.1 Polarity. The left-hand or upper terminal (with the ammeter in the normal mounting position and viewed from the rear) shall be the positive terminal and shall be so marked.

3.5.8.2 Bonding, electrical. All permanent magnet, moving coil ammeters shall have the magnetic system, metallic dial and internal metallic bezel ring (if used and insulated from case) electrically connected to the negative terminal.

3.6 Mounting hardware. Machine screws, nuts, split-ring lockwashers, and bolts or studs shall be furnished with each ammeter for use in mounting. The machine screws shall have a semigloss gray finish. Mounting shall be by the flanges only and use shall not be made of any supplementing brackets behind the panel (see 3.11).

3 7 Self-contained limitations. Ammeters shall be self contained for ranges of 500 milliamperes, or less.

3.8 Internal illumination. When specified (see 1.2.1.8 and 6.1), ammeters shall be equipped with internal illumination provided by two 6 volt N.328 incandescent lamps in accordance with MS 25237 and placed so as to illuminate the scale. The lamps shall be readily replaceable from behind the mounting panel. The internal illumination feature shall be used only with the black background, white translucent marking scales. Each ammeter so illuminated shall be fitted with a red filter which limits light transmitted through the ammeter window to wave lengths above 6000 angstrom units. These filters may be part of the scale. The lamp circuit shall be electrically isolated from the case. Light emitted by any portion of the internally illuminated scale shall not exceed 1 foot-lambert nor be less than 0.2 foot-lambert.

3.9 Performance requirements. Ammeters shall conform to the requirements of USAS C39.1-1964 for edgewise panel type, direct current, classes 2 and 3, as applicable, for case sizes of 5 and 8 inches. The exceptions and additional requirements specified in 3.9.1 through 3.9.7 shall apply.

3.9.1 Accuracy. Rated accuracy of the ammeters shall be 1.5 percent of full scale value.

3.9.2 Vibration. When tested as specified in 4.5.2, ammeters shall show no evidence of breakage, permanent deformation, or loosening of parts and shall retain their serviceability. After this test, the friction error shall not exceed 1 percent of scale length, and the permanent change in indication shall not exceed 2 percent of full scale value (see 3.2.1).

3.9.3 Shock. When tested as specified in 4.5.1, ammeters shall show no evidence of breakage, permanent deformation or loosening of parts, and shall retain their serviceability. After this test, the permanent change in indication shall not exceed 7 percent of scale length.

3.9.4 External-field influence. An external magnetic field of 5 oersteds shall not cause a change in indication exceeding 1.5 percent of scale length.

3.9.5 Dripproofness. Ammeter windows and flanges shall effectively exclude entry of water into the ammeter case or past the mounting panel when a mounting panel containing an ammeter is subjected to the dripproof test (45° degrees) of MIL-STD-108. This requirement applies only to the ammeter and panel front (see 4.6.4).

3.9.6 Overload.

3.9.6.1 Momentary overload. Ammeter circuits shall withstand nine applications of 10 times full scale current for 0.5 second with 1 minute intervals between applications, followed by one application of the same current for 5 seconds. Following zero reset as required, change in indication shall not exceed 2 percent of scale length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:31Z
Check the source to verify that this is the current version before use.

3.9.6.2 Sustained overload. Ammeter circuits shall withstand application of 120 percent full scale current for 8 hours. Following a 16 hours storage period at room temperature and zero reset as required, the permanent change in indication shall not exceed 1 percent of scale length.

3.9.7 Humidity resistance. After the tests specified in 4.3, ammeters shall operate freely and the permanent change in indication shall not exceed 3 percent of scale length.

3.10 Marking.

3.10.1 Scale. In addition to the quantity measured, and the scale divisions, each ammeter scale shall be marked with the manufacturer's trademark.

3.10.2 Identification plate. A metal of plastic identification plate shall be bonded or otherwise affixed to the ammeter body behind the mounting flange. The plate shall contain information as follows:

(a) Manufacturer's trademark or name.
(b) Quantity measured by ammeter, d c. milliamperes.
(c) Full scale deflection of ammeter (referred to meter terminals, irrespective of any external accessories needed).
(d) Loss:

1. Expressed in millivolts for d.c. current measuring instruments (milliammeters).

(e) Type designation (see 1.2.1).

3.11 Size, mounting, and terminal placement Size, mounting panel cutout, mounting hardware and holes, and terminal placement and hardware shall be as show on figure 1 for 5-inch and figure 2 for 8-inch ammeters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 Classification of inspections. Inspections shall be classified as follows.

(a) First article inspection
(b) Quality conformance inspection

4.3 First Article inspection. Three sample ammeters of each size shall be subjected to all of the tests shown in table I. The rating for first article inspection shall be o to 20 milliamperes.

Table I - First article inspection

Test	Applicable Requirements Paragraph	Applicable Test Paragraph
Visual and mechanical examination (Groups I and II)		4.6.3
Rated accuracy	3.9.1	USAS C39.1
Damping factor	--	USAS C39.1
Response time	USAS C39.1	USAS C39.1
Loss	USAS C39.1	USAS C39.1
Position influence	USAS C39.1	USAS C39.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:31Z
Check the source to verify that this is the current version before use.

Table I - First article inspection (Cont'd.)

Test	Applicable Requirement Paragraph	Applicable Test Paragraph
Extreme temperature	USAS C39.1	USAS C39.1
External temperature	USAS C39.1	USAS C39.1
External field influence	3.9.4	4.6.5
Overload	3.9.6	USAS C39.1
Vibration	3.9.2	4.6.2
Shock (2 samples of each size only)	3.9.3	4.6.1
Enclosure (1 sample of each size only)	3.9.5	4.6.4
Dielectric test	USAS C39.1	USAS C39.1
Leakage current	USAS C39.1	USAS C39.1
Proximity influence	USAS C39.1	USAS C39.1
Humidity resistance	3.9.7	USAS C39.1
Internal illumination	3.8	4.6.6

4.4 Production control inspection. The tests shown in table I shall be performed following any design change or change of any material (of a vital part) and at three year intervals thereafter.

4.5 Production inspection. Every ammeter offered for delivery shall be subjected to the test in table II to determine conformance with this specification.

Table II - Production inspection.

Test	Applicable Requirement Paragraph	Applicable Test Paragraph
Visual and mechanical examination (Group II only)		4.6.3
Rated accuracy	3.9.1	USAS C39.1
Position influence	USAS C39.1	USAS C39.1
Dielectric test	USAS C39.1	USAS C39.1
Leakage current	USAS C39.1	USAS C39.1

4.6 Inspection details.

4.6.1 Shock. Two of the three ammeters of each size tested in 4.3 shall be subjected to high-impact shock tests in accordance with type C of MIL-S-901 for lightweight equipment when mounted on fixture 6D-1 as shown on figure 7-1 of MIL-S-901. During the test, one sample shall be energized to approximately the center scale value. The other sample shall be deenergized. Wiring connected to the energized sample shall be firmly attached to the panel at a point 18 inches or less from the sample. The pointer shall be reset to zero after each blow. Sequence of shock blows shall be back, top, side. At the completion of the shock test, the meters shall be checked for error of indication, then disassembled and examined for damage.

4.6.2 Vibration. Ammeters tested in 4.6.1 shall be subjected to vibration in accordance with type I of MIL-STD-167. The change in indication and friction error shall be determined, after reset to zero, if required, and the meter shall be examined for evidence of breakage, permanent deformation or loosening of parts.

4.6.3 Visual and mechanical examination. Visual and mechanical examination shall be performed in accordance with table III to determine conformance with this specification.

Table III - Visual and Mechanical examination

Examination	Applicable Requirements Paragraph
GROUP I:	
Materials	3.3, 3.5
Scale length	3.9
Mounting	3.6, 3.11

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

Table III - Visual and Mechanical examination (cont'd.)

Examination	Applicable Requirement Paragraph
GROUP I: (cont'd.)	
Workmanship (with disassembly)	Meter examined for dust or other contaminations, including loose flakes or particles due to shock test or corrosion
Internal illumination	3.8
GROUP II.	
Marking, scale	3.5.6.2
Marking, case	3.10.2
Marking, terminals	3.5.8.1.1
Workmanship (without dissassembly)	sound practice

4.6.4 Dripproofness. One of the ammeters of each size tested in 4.3 which has not been subjected to the test of 4.6.1 shall be subjected to the test specified in MIL-STD-108 for dripproof (45 degree) enclosures. The meter shall be mounted in the normal manner in a panel and the front surface only shall be subjected to this test. Entrance of any liquid into the meter case (as determined by disassembly and inspection) shall be cause for rejection.

4.6.5 External field influence. The ammeter shall be placed at the center of a coil carrying direct current. The coil shall carry a current sufficient to produce a uniform field of approximately 5 oersteds. Readings shall be taken with the ammeter at various positions in the center of the field to determine conformance with 3.9.4. An acceptable configuration is a coil of four turns 50 centimeters in diameter, carrying a current of 50 amperes. Other coil sizes and currents are acceptable if an equivalent electrical field is produced.

4.6.6 Internal illumination. Ammeter manufacturer shall provide Objective Quality Evidence that color of light emitted from ammeters conforms with 3.8. Light intensity shall be determined by positioning a suitable sensitive light meter on the extended ammeter centerline at a distance of 1 foot from the window.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation

5.1.1 Basic equipment or item.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

(5.2 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.)

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

5.2.1 Preservation and packaging (see 6.1).

5.2.1.1 Level A. Ammeters and mounting hardware shall be individually protected and unit-packaged in accordance with method III of MIL-P-116. Prior to packaging, all ammeters shall be suitable damped as necessary utilizing a wire or other shunt.

5.2.1.2 Level C. Ammeters shall be afforded preservation and packaging in accordance with the supplier's normal commercial practice.

5.2.2 Packing (see 6.1).

5.2.2.1 Level A. Ammeters packaged as specified (see 6.1) shall be packed in overseas type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 2 or 3, as specified (see 6.1)), or wood-cleated paper-overlaid boxes conforming to PPP-B-576, PPP-B-585, PP-B-591, PP-B-601, PPP-B-621, and PPP-B-636, respectively, at the option of the supplier. Shipping containers shall have case liners conforming to MIL-L-10547, the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636 may be omitted provided the center and edge seams and manufacturer's joints are sealed with tape, at least 1-1/2 inches wide, conforming to PPP-T-76. Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to PPP-B-636 may be banded with tape conforming to type IV of PPP-T-97 and appendix thereto in lieu of steel straps The gross weight of wood boxes shall not exceed 200 pounds, fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2.2 Level B. Ammeters packaged as specified (see 6.1) shall be packed in domestic type wirebound wood, wood-cleated fiberboard, wood-cleated, plywood, nailed wood, fiber (class 1 or 2, as specified (see 6.1)), or wood-cleated paper-overlaid boxes conforming to PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, and PPP-B-636, respectively, at the option of the supplier. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds, fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2.3 Level C. Ammeters packaged as specified (see 6.1) shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2.3 General Insofar as possible and practical, exterior containers shall be uniform in shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities of identical items.

5.2.4 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with MIL-STD-129 (see 6.1)).

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows.

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

Source: https://assist.dla.mil -- Downloaded: 202640-19T21:31Z
Check the source to verify that this is the current version before use.

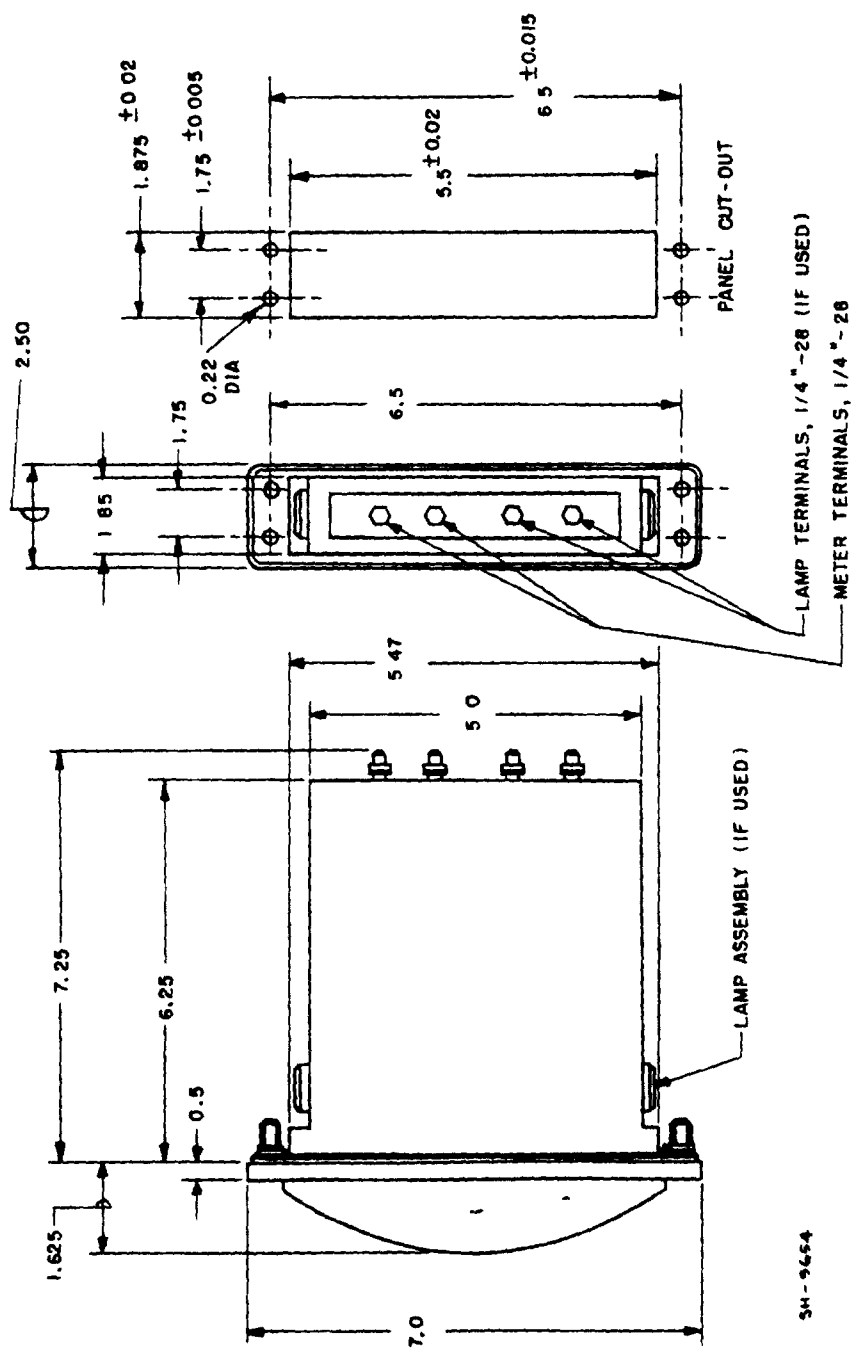
6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type designation (see 1.2.1).
(c) Description of special scale markings, when specified (see 1.2.1.9).
(d) Sample for first article inspection (see 3.1).
(e) Internal illumination, when specified (see 3.8).
(f) Level of preservation, packaging, packing and marking (see 5.2).
(g) Approval of polystyrene (loose-fill) material (see 5.3).

Preparing activity:
Navy - SH
(Project 6625-N263)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:31Z
Check the source to verify that this is the current version before use.

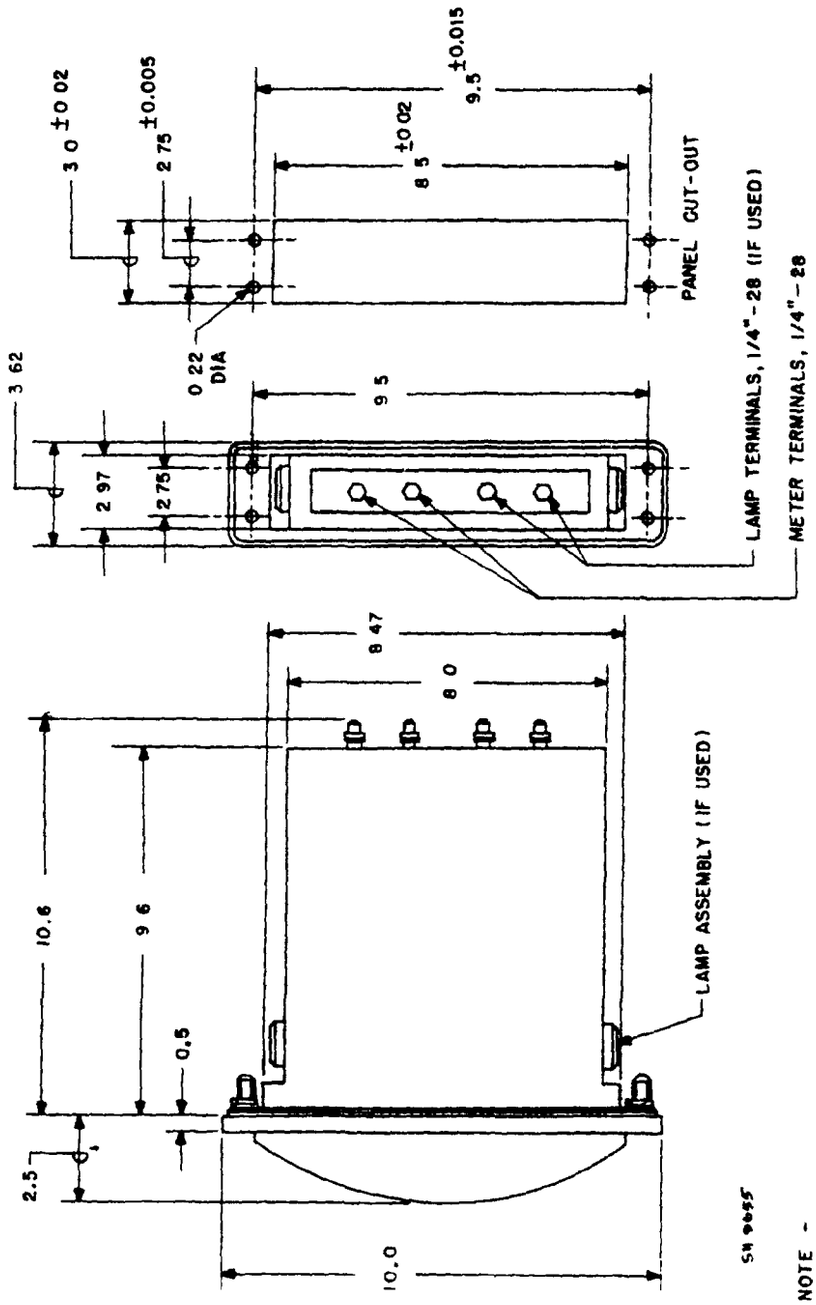


NOTE.
ALL DIMENSIONS MAXIMUM (UNLESS OTHERWISE NOTED)

FIGURE-1 DESIGN, CONSTRUCTION AND PHYSICAL DIMENSIONS FOR 5-INCH AMMETER

NOT TO SCALE

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.



NOTE -
ALL DIMENSIONS MAXIMUM (UNLESS OTHERWISE NOTED)

FIGURE-2 DESIGN, CONSTRUCTION AND PHYSICAL DIMENSIONS FOR 8 INCH AMMETER

NOT TO SCALE

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1426 REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE 615886
1 APR 65 (NAVSHIPS OVPRT 13-66)

Source: https://assist.dla.mil -- Downloaded: 202610-19T21:31Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL SHIP ENGINEERING CENTER
WASHINGTON, D C 20360

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

COMMANDER NAVAL SHIP ENGINEERING CENTER
DOD STANDARDIZATION PROGRAM AND DOCUMENTS BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D C 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-19T21:31Z
Check the source to verify that this is the current version before use.